Assignment 4 - with responses

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Packages

```r
library(tidyverse)   # Graphs with ggplot()
library(knitr)       # Tables with kable()
library(ggokabeito)  # colorblind friendly colors
library(viridis)     # colorblind friendly colors
library(brms)        # Hierarchical Bayesian models
library(rstan)       # Interface to Stan
library(shinystan)   # Model output graphs
library(bayesplot)   # Model output graphs

# General settings
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)
options(contrasts = c("contr.equalprior_pairs", "contr.poly"))
set.seed(1234) # For reproducibility
```

Preparation

Data: `penguins`

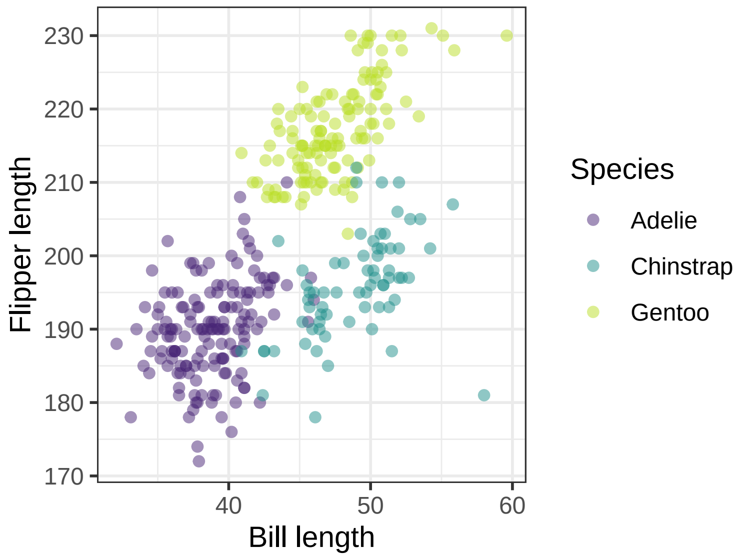
```r
head(penguins)
```

```
##   species    island bill_len bill_dep flipper_len body_mass    sex year
## 1  Adelie Torgersen     39.1     18.7         181      3750   male 2007
## 2  Adelie Torgersen     39.5     17.4         186      3800 female 2007
## 3  Adelie Torgersen     40.3     18.0         195      3250 female 2007
## 4  Adelie Torgersen       NA       NA          NA        NA   <NA> 2007
## 5  Adelie Torgersen     36.7     19.3         193      3450 female 2007
## 6  Adelie Torgersen     39.3     20.6         190      3650   male 2007
```

The data consist of several measurements of penguins of the Palmer archipel in Antarctica.

Variable	Description
`species`	Factor: Adelie, Chinstrap, Gentoo
`island`	Factor: Biscoe, Dream, Torgensen
`bill_len`	Numeric: Mean = 43.91, range [32.10, 59.6]
`bill_dep`	Numeric: Mean = 17.15, range [13.10, 21.5]
`flipper_len`	Numeric: Mean = 200.9, range [172, 231]
`body_mass`	Numeric: Mean = 4202, range [2700, 6300]
`sex`	Factor: female, male
`year`	Numeric: 2007, 2008, 2009

```r
penguins <- penguins
g1 <- ggplot(penguins, aes(x = bill_len, y = flipper_len, color = species))+
  geom_point(alpha = .5)+
  scale_color_viridis_d(begin = .1, end = .9)+
  labs(x = "Bill length", y = "Flipper length", color = "Species")+
  theme_bw()
g1
```




```r
# Remove missing values
penguins <- na.omit(penguins)

# Center numeric predictors in case you build models with interaction terms
penguins$bill_len_c <- penguins$bill_len - mean(penguins$bill_len)
penguins$bill_dep_c <- penguins$bill_dep - mean(penguins$bill_dep)
```

Excercises

Analyse the penguins dataset to investigate the hypothesis that flipper length is influenced by bill length and bill depth (at this point without considering interactions). Do so following a series of steps in the Bayesian workflow.

Exercise 1.

Translate the hypothesis into a Bayesian mathematical model. What is the general structure of this model?

Assume Y_i is the flipper length of the ith penguin, X_{i1} is the bill length of the ith penguin, and X_{i2} is the bill depth of the ith penguin. Then

$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$

$\mu_i = \beta_0 + X_{i1}\beta_1 + X_{i2}\beta_2$

We need priors on all β and σ.

Exercise 2.

Take a look at the default priors for the model, and define two sets of alternative priors. Hint: You can use the `default_prior()` function to see the default prior structure for this model.

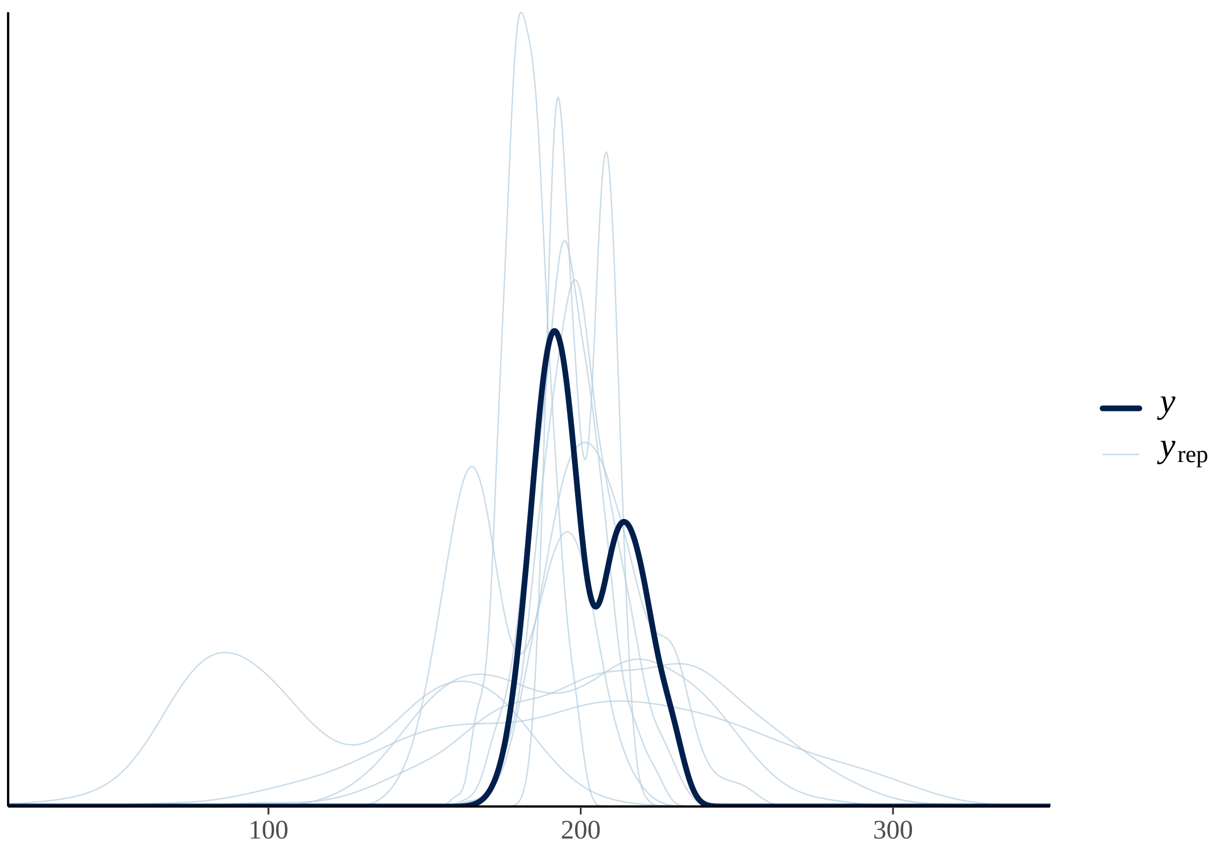
```
# Take a look at the default priors and the
# corresponding names of each type of parameter
default_prior(
  flipper_len ~ 1 + bill_len + bill_dep
  , data = penguins
)

##                    prior     class     coef group resp dpar nlpar lb ub tag
##                   (flat)         b
##                   (flat)         b bill_dep
##                   (flat)         b bill_len
##  student_t(3, 197, 16.3) Intercept
##    student_t(3, 0, 16.3)     sigma                                 0
##        source
##       default
##  (vectorized)
##  (vectorized)
##       default
##       default

# This model has parameters of three types:
# b, Intercept, and sigma
# There are two parameters of type b, one for each predictor
# There is one parameter for sigma (with a lower bound of 0)
# and one parameter for the intercept

# Alternative priors for fixed predictors (b)
altPrior1 <- c(prior(normal(0, 4), class = b, coef = bill_dep)
             ,(prior(normal(0, 5), class = b, coef = bill_len)
            ))
# These priors are a bit more informativ than the default flat priors.
# The priors leave open whether there is a postiive, negative, or null correlation
# between the predictors (bill measurements) and the criterion (flipper length).
# They also assume the penguins have bills that are longer than they are wide
# and that therefore the effect of bill_len can be expected to be stronger than
# the effect of bill_dep.

# Alternative priors for predictos (b) and variance (sigma)
altPrior2 <- c(prior(normal(0, 4), class = b, coef = bill_dep)
             , prior(normal(0, 5), class = b, coef = bill_len)
             , prior(normal(0, 10), class = sigma)
            )
# This pior variance is similar to the default but slightly more informative
```

Exercise 3.

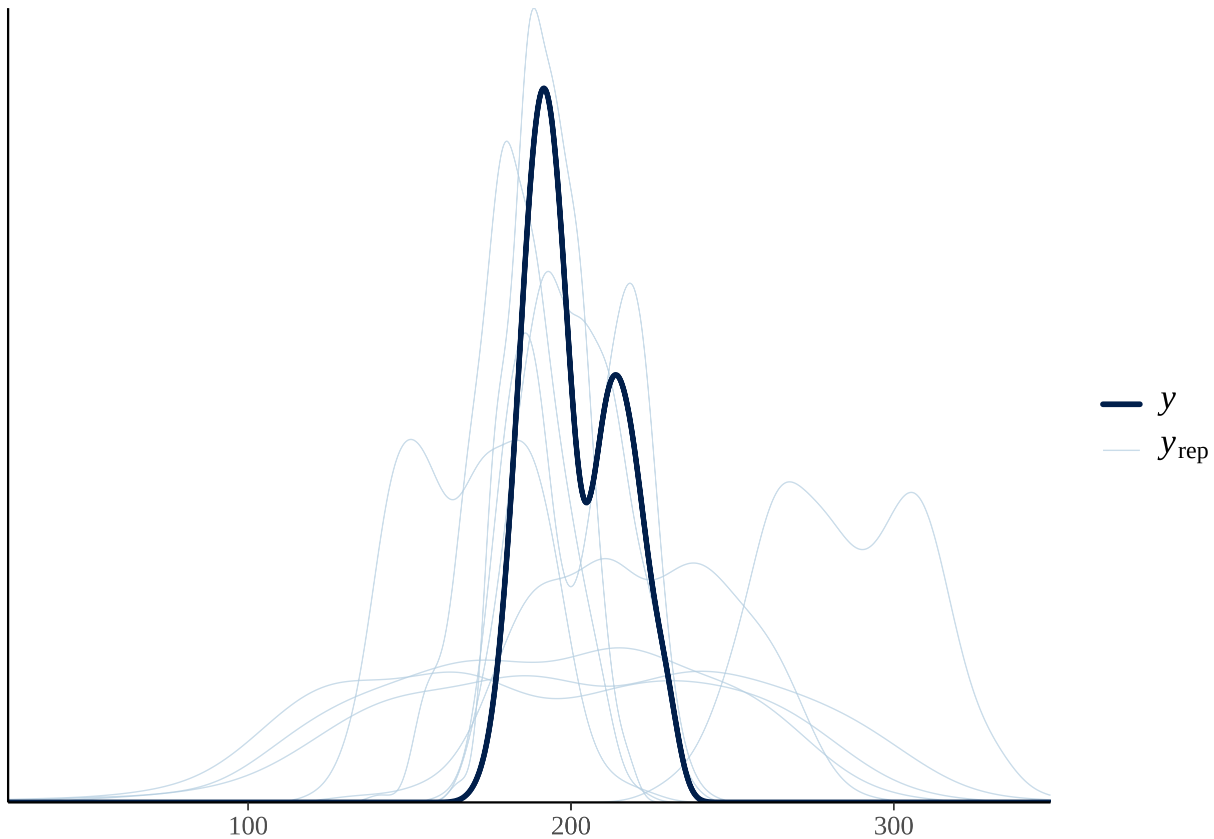
Conduct a small sensitivity analysis by comparing the prior predictive values of the model for the three sets of priors.

```
## output <- capture.output(
## defaultprior_pred <- brm(flipper_len ~ 1 + bill_len + bill_dep
##                   , data = penguins
##                   , sample_prior = "only"
##                   , iter = 1000
## )
##)
```

```r
# No prior predictive values can be obtained when the priors are flat
# Flat priors are "improper", i.e. they violate the axioms of probability theory.
# This is a reason for choosing proper priors instead.

output <- capture.output(
 altPrior1_pred <- brm(flipper_len ~ 1 + bill_len + bill_dep
                  , data = penguins
                  , prior = altPrior1
                  , sample_prior = "only"
                  , iter = 1000
 )
)

output <- capture.output(
 altPrior2_pred <- brm(flipper_len ~ 1 + bill_len + bill_dep
                  , data = penguins
                  , prior = altPrior2
                  , sample_prior = "only"
                  , iter = 1000
 )
)

# plot(altPrior1_pred) Take a look at the simulation output
prior_predictions1 <- pp_check(altPrior1_pred)
prior_predictions2 <- pp_check(altPrior2_pred)
prior_predictions1
```




```r
prior_predictions2
```



Exercise 4.

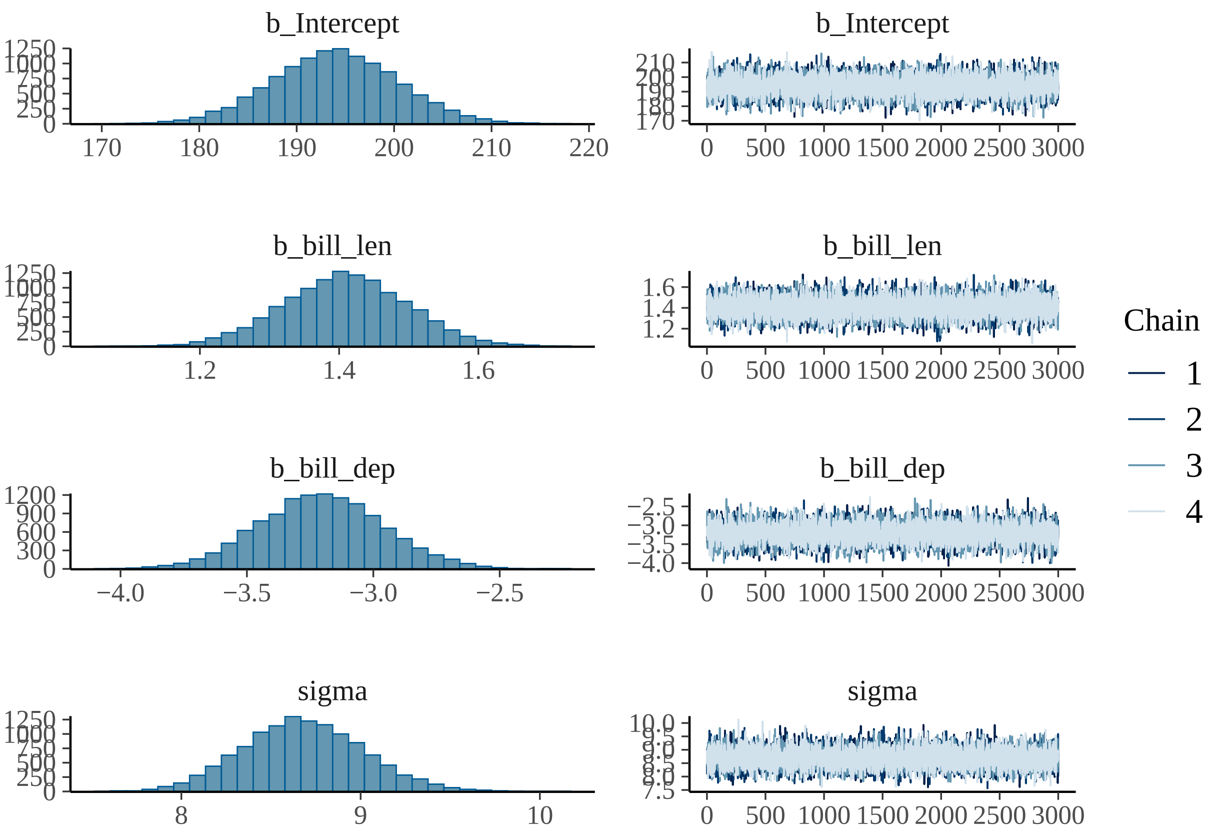
Apply the model to the data with `brm()`, using the priors that faired best in the prior predictive checks. Assess the convergence parameters and briefly interpet the model output.

```
output <- capture.output(
m1 <- brm(
  flipper_len ~ bill_len + bill_dep
  , data = penguins
  , prior = altPrior2
  , iter = 4000
  , warmup = 1000
  , chains = 4
    , cores = 4
    , save_pars = save_pars(all = TRUE)
   )
)

# Summary statistics for the model output
# and convergene diagnostics
summary(m1, waic = TRUE)
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: flipper_len ~ bill_len + bill_dep
##    Data: penguins (Number of observations: 333)
##   Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
##          total post-warmup draws = 12000
##
```

```
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept   194.00      6.45   181.28   206.65 1.00    10998     8996
## bill_len      1.41      0.09     1.23     1.58 1.00    12395     9258
## bill_dep     -3.21      0.25    -3.69    -2.71 1.00    12146     9054
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     8.68      0.34     8.05     9.36 1.00    12840     9893
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m1)
```




```r
m1$fit
```

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=4000; warmup=1000; thin=1;
## post-warmup draws per chain=3000, total post-warmup draws=12000.
##
##                mean se_mean   sd    2.5%     25%     50%     75%   97.5%
## b_Intercept  194.00    0.06 6.45  181.28  189.62  194.02  198.36  206.65
## b_bill_len     1.41    0.00 0.09    1.23    1.35    1.41    1.47    1.58
## b_bill_dep    -3.21    0.00 0.25   -3.69   -3.37   -3.20   -3.04   -2.71
## sigma          8.68    0.00 0.34    8.05    8.45    8.67    8.91    9.36
## Intercept    200.97    0.00 0.48  200.03  200.65  200.97  201.28  201.90
## lprior       -11.93    0.00 0.06  -12.05  -11.97  -11.93  -11.89  -11.83
```

```
## lp__        -1201.74    0.02 1.41 -1205.36 -1202.42 -1201.42 -1200.71 -1199.98
##             n_eff Rhat
## b_Intercept 11013    1
## b_bill_len  12481    1
## b_bill_dep  12139    1
## sigma       12844    1
## Intercept   13008    1
## lprior      12454    1
## lp__         6327    1
##
## Samples were drawn using NUTS(diag_e) at Sun Aug 23 22:47:41 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

```
# See also: launch_shinystan(m1)
```

The effective sample sizes for each parameter are above 1000, and the Gelmann-Rubin criterion $\hat{R}$ ("R hat") is not larger than 1. The four chains have mixed well, showing a horizontal zig-zag with no discernable systematic drift upwards or downwards. The posterior distributions for each parameter are unimodal and relatively smooth; and they occupy limited regions of the parameter space rather than stretching across all or most possible parameter values. All this speaks in favour of the model having converged to a stable solution.

None of the credible intervals include the Null. `bill_len` seems to have a slightly positive effect, and `bill_dep` a slightly negative effect on `flipper_len`.

Exercise 5.

Define two alternative models: An intercept-only model without predictors, and a complexer model that includes not just the main effects of `bill_len` and `bill_dep`, but also their interaction. Compare these models using Bayes factors, and briefly interpret the results.

```
# Because the intercept-only has no predictors,
# we need to adjust the prior
default_prior(
  flipper_len ~ 1
  , data = penguins
)
```

```
##                    prior     class coef group resp dpar nlpar lb ub tag  source
##  student_t(3, 197, 16.3) Intercept                                      default
##    student_t(3, 0, 16.3)     sigma                             0        default
```

```
altPrior0 <- prior(normal(0, 10), class = sigma)

output <- capture.output(
m0 <- brm(
  flipper_len ~ 1
  , data = penguins
  , prior = altPrior0
  , iter = 4000
  , warmup = 1000
  , chains = 4
    , cores = 4
    , save_pars = save_pars(all = TRUE)
   )
)
```

```r
output <- capture.output(
m2 <- brm(
  flipper_len ~ bill_len * bill_dep
  , data = penguins
  , prior = altPrior2
  , iter = 4000
  , warmup = 1000
  , chains = 4
    , cores = 4
    , save_pars = save_pars(all = TRUE)
   )
)
```

```r
summary(m0, waic = TRUE)
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: flipper_len ~ 1
##    Data: penguins (Number of observations: 333)
##   Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
##          total post-warmup draws = 12000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept   200.95      0.77   199.43   202.47 1.00    11425     7910
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma    14.03      0.55    13.01    15.13 1.00     9339     7260
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
summary(m2, waic = TRUE)
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: flipper_len ~ bill_len * bill_dep
##    Data: penguins (Number of observations: 333)
##   Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
##          total post-warmup draws = 12000
##
## Regression Coefficients:
##                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept           -45.08     36.48  -116.94    26.94 1.00     3651     4656
## bill_len              6.71      0.80     5.11     8.30 1.00     3610     4712
## bill_dep             10.26      2.04     6.21    14.27 1.00     3635     4538
## bill_len:bill_dep    -0.30      0.05    -0.39    -0.21 1.00     3609     4609
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     8.08      0.32     7.49     8.72 1.00     6538     6207
##
```

```
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
# The bayes_factor() function uses bridge sampling
bf_10 <- bayes_factor(m1, m0)
```

```
## Iteration: 1
## Iteration: 2
## Iteration: 3
## Iteration: 4
## Iteration: 1
## Iteration: 2
## Iteration: 3
## Iteration: 4
```

```
bf_10
```

```
## Estimated Bayes factor in favor of m1 over m0: 542540924537438272635897995843270079255583787890620649
```

```
bf_21 <- bayes_factor(m2, m1)
```

```
## Iteration: 1
## Iteration: 2
## Iteration: 3
## Iteration: 4
## Iteration: 5
## Iteration: 1
## Iteration: 2
## Iteration: 3
## Iteration: 4
## Iteration: 5
```

```
bf_21
```

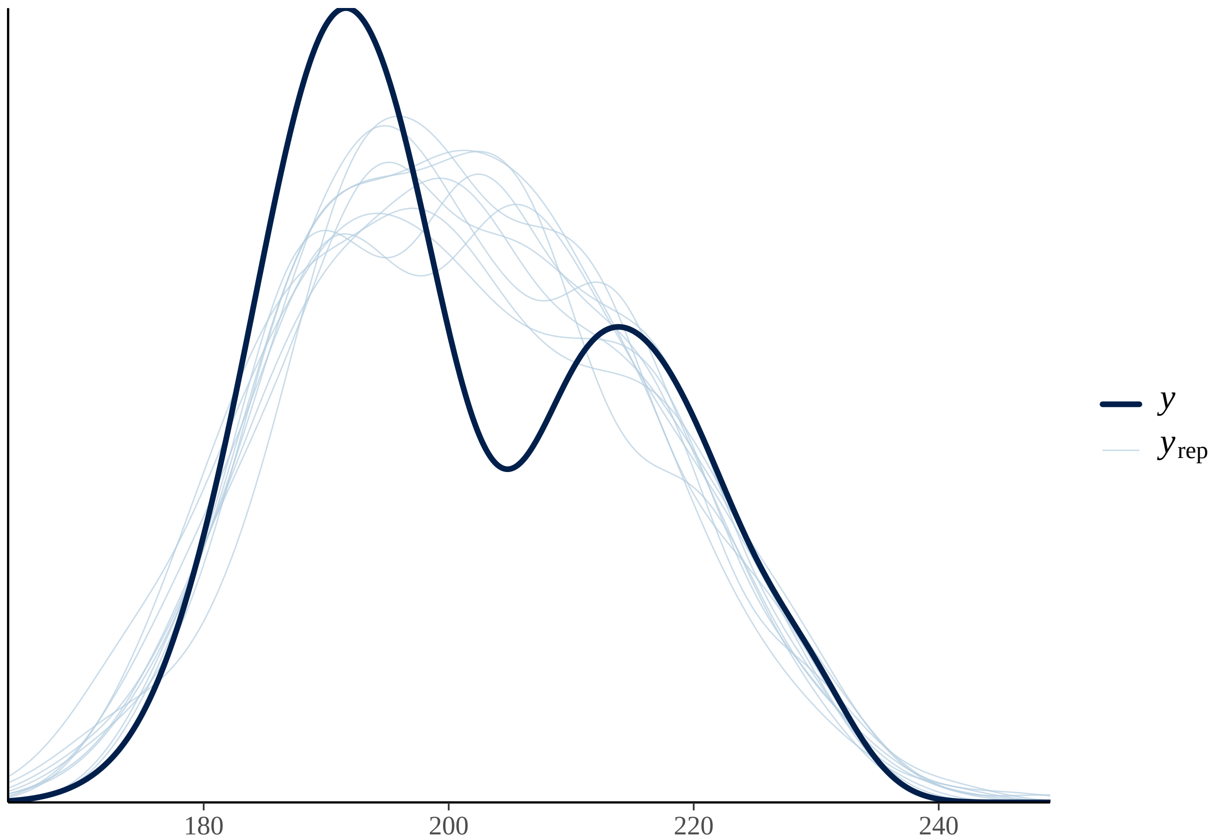
```
## Estimated Bayes factor in favor of m2 over m1: 87658915.93862
```

There is very strong evidence (BF > 100) that model `m2` can account for the pattern in the data better than `m1`, which in turn can account for the data better than model `m0`.

Exercise 6

Conduct posterior predictive checks for the model that fairs best on the basis of the Bayes factor comparison.

```
pp_check(m2)
```


180
200
220
240
y
y rep